$$\begin{aligned}\frac{x+1-3(x+3)}{x+3} &< 0\\ \frac{-2x-8}{x+3} &< 0\\ \frac{x+4}{x+3} &> 0\end{aligned}$$

Thus, $x+4$ and $x+3$ are both positive or both negative.

So let us find out the signs of $x+3$ and $x+4$ as follows

x	$x+3$	$x+4$	$\frac{x+4}{x+3}$
$x<-4$	$-$	$-$	$+$
$-4<x<-3$	$-$	$+$	$-$
$x>-3$	$+$	$+$	$+$
$x=-4$	$-$	0	0

So the solution set is given by $(-\infty,-4)\cup(-3,\infty)$.

The above type of rational inequality problem can also be solved by plotting the signs of various factors on the intervals of the number line.

Exercise - 2.8

1. Find all values of x for which $\dfrac{x^3(x-1)}{(x-2)}>0$.
2. Find all values of x that satisfies the inequality $\dfrac{2x-3}{(x-2)(x-4)}<0$.
3. Solve $\dfrac{x^2-4}{x^2-2x-15}\leq 0$.

2.7.2 Partial Fractions

A rational expression $\dfrac{f(x)}{g(x)}$ is called a proper fraction if the degree of $f(x)$ is less than degree of $g(x)$, where $g(x)$ can be factored into linear factors and quadratic factors without real zeros. Now $\frac{f(x)}{g(x)}$ can be expressed in simpler terms, namely, as a sum of expressions of the form

(i) $\dfrac{A_1}{(x-a)}+\dfrac{A_2}{(x-a)^2}+\cdots+\dfrac{A_k}{(x-a)^k}$ if $x-a$ divides $g(x)$ and

(ii) $\dfrac{(B_1x+C_1)}{(x^2+ax+b)}+\dfrac{(B_2x+C_2)}{(x^2+ax+b)^2}+\cdots+\dfrac{(B_kx+C_k)}{(x^2+ax+b)^k}$ if x^2+ax+b has no real zeros and (x^2+ax+b) divides $g(x)$.

The resulting expression of $\dfrac{f(x)}{g(x)}$ is called the *partial fraction decomposition*. Such a decomposition is *unique* for a given rational function.

This method is useful in doing Integral calculus. So let us discuss some examples.